Customer churn: who is about to leave, and what to do

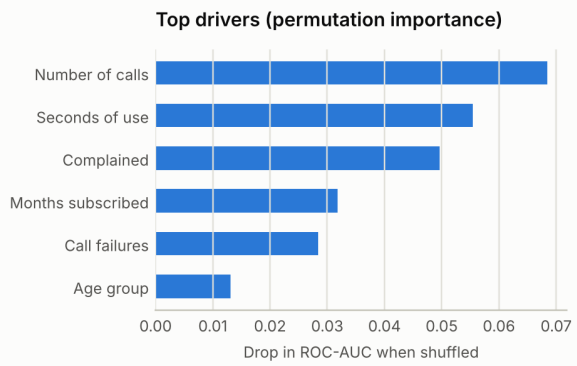
One-page summary of a predictive model · telecom operator, 2,850 unique customers · Sample project built with public data

0.984
ROC-AUC on unseen customers (1.0 = perfect)

79%
of churners caught (88 of 112)

91%
of flagged customers really churn (88 of 97)

14%
of customers flagged for a retention call



What the model does. Gives each customer a probability of leaving in the next quarter, from 9 months of usage data; it can re-score the customer base every month. A gradient-boosting model beat a random forest and a logistic regression in 5-fold cross-validation (PR-AUC 0.95). It was tested once on 713 customers it had never seen. Customers above the cut-off (0.85) go on the retention list; the cut-off was chosen on training data to catch about 80% of churners.

Who leaves. The lowest-usage quarter of customers churns at 35%, the highest at 0%. Customers who complained churn at 83% (vs 10%).

Five recommendations (based on the model; test each before rolling it out)

1. **Treat every complaint as a churn alarm.** 83% of customers who complained left, against 10% of the rest, and complaints are a top-3 driver. Route each complaint to a retention agent and confirm it was solved.
2. **Watch for falling usage every month.** Number of calls and seconds of use are the two strongest drivers. A drop in usage is the earliest signal the company sees: trigger contact when a customer's monthly usage falls well below their own average.
3. **Re-activate non-active customers.** 48% of customers with non-active status churned, against 6% of active ones. A usage-based incentive is a natural test.
4. **Spend the retention budget on the model's list.** Flagging 14% of customers catches 79% of the churners, and 91% of the flagged customers are real churners, so few offers are wasted. Call the highest scores first.
5. **Test contract plans with a control group.** Contract customers churned at 3% vs 17% on pay-as-you-go, but the model gives the plan little weight once usage is known. Offer migration to a random half of at-risk pay-as-you-go customers and compare.

Data: Iranian Churn Dataset, UCI Machine Learning Repository, DOI 10.24432/C5JW3Z, CC BY 4.0. Features cover months 1-9, churn is measured at month 12. 300 duplicate rows removed before the train/test split to avoid leakage. **Limits:** one operator and period; no revenue or offer-cost data, so the cut-off is set by a recall target; associations, not causes. Full notebook: analysis/churn_model.ipynb.